

Ranhui Yan

Teaching Assistant and Machine Learning Researcher

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Research Profile

My research interests lie in graph learning and graph neural networks, with a particular focus on spectral methods and robustness. My current work studies the theoretical properties of spectral graph neural networks and their robustness to structural perturbations. I am also interested in extending graph learning methods to scientific and interdisciplinary problems.

Research Interests

- Graph learning and graph neural networks
- Spectral graph neural networks and their theoretical properties
- Robustness to structural perturbations in graph learning
- Graph learning for scientific and interdisciplinary applications

Experience

Teaching Assistant

2024–Present

School of Computer and Artificial Intelligence, Guangzhou Xinhua University, Guangzhou, China

- Teach and support courses in Calculus, Linear Algebra, Statistics, and Machine Learning.

Education

M.S. in Statistics

2021–2024

Guangdong University of Finance & Economics, Guangzhou, China
Supervised by Prof. Jia Cai.

B.S. in Information and Computing Science

2017–2021

China Jiliang University, Hangzhou, China

Publications

R. Yan, J. Cai, M. Chen, and H. Yang. "Convex–Concave Quadratic Spectral Filtering for Graph Neural Networks." *Accepted at ECML 2026*. [arXiv:2606.24956](#). [Code](#).

R. Yan and J. Cai. "Virtual Nodes based Heterogeneous Graph Convolutional Neural Network for Efficient Long-Range Information Aggregation." *International Conference on Artificial Neural Networks (ICANN)*, 2024, pp. 217–231. [arXiv:2509.23660](#). [DOI](#). [Code](#).

Z. Xiong, J. Cai, **R. Yan**, and X. Huang. "Noise Perturbation Based Graph Contrastive Learning via Flexible Filters for Node Classification." *International Joint Conference on Neural Networks (IJCNN)*, 2024, pp. 1–10. [DOI](#).

R. Yan and J. Cai. "An End-to-End Dense Connected Heterogeneous Graph Convolutional Neural Network." *International Conference on Neural Information Processing*, 2023, pp. 462–475. [DOI](#).